

Designing a Dual-Resource Progression Economy in an Asynchronous Game

Introduction

Designing a fair and engaging progression economy in an asynchronous, user-generated content game requires carefully balancing multiple resource loops and player motivations. In the hypothetical **Verdant Coil** scenario, players both **create levels** and **play through others' levels** in 3–10 minute loops. Two distinct resources drive progression: **Sporeprint** (earned when other players complete your level, used for avatar upgrades) and **Nutrient** (earned by sending your “crawler” through others' levels, used to expand your Coil/base). This dual-resource system resembles mechanics in games like *The Mighty Quest for Epic Loot*, where players build trap-filled dungeons and raid others' dungeons, yielding separate currencies for defense vs. offense ¹. To ensure long-term engagement in a non-commercial, education-driven project, the economy must prioritize **fairness, fun, and ongoing engagement** over monetization. Key design challenges include balancing the two resources, setting appropriate reward structures (static vs. dynamic), preventing inflation or farming exploits, providing meaningful “sinks” for high-level players, smoothing progression for beginners while preserving challenge for veterans, and leveraging psychological motivators (progress feedback, variety, mastery, collection) to enhance enjoyment.

Below, we break down best practices and potential failure modes for each of these aspects, drawing on game design literature, industry insights (e.g. GDC talks, whitepapers), and analogies from comparable games with persistent UGC (user-generated content) ecosystems.

Balancing a Dual-Resource Economy (Sporeprint vs. Nutrient)

Introducing two interdependent resources can enrich gameplay by encouraging players to engage in **multiple roles or activities**. The core idea is to tie each currency to a different gameplay loop so neither creators nor players can ignore one side of the game. For example, *Clash of Clans* uses two main resources (gold and elixir) which serve distinct purposes (building defenses vs. training troops); this ensures players invest in both base-building and raiding, rather than focusing on a single optimal strategy. In **Verdant Coil**, Sporeprint rewards the **creator role** (designing levels) while Nutrient rewards the **player/solver role** (beating levels). Key considerations for balancing these are:

- **Earning Rates & Interdependency:** If one resource is significantly easier to earn or dramatically more needed than the other, the game risks imbalance. Players may ignore the “hard” resource's activity or feel forced into unfun grinding. A well-tuned system offers a **comparable progression rate** for both creation and play. *Mighty Quest for Epic Loot* balanced this by allowing non-paying players to **grind both gold and life-force at reasonable rates**, either by waiting for their own castle's resources to accumulate or by raiding others – both paths required time but neither was overwhelmingly more efficient ¹. Designers should similarly ensure that running your crawler (to earn Nutrient) and hosting levels (to earn Sporeprint) are **equally rewarding over time**.

- **Cross-Utilization vs. Segregation:** Decide if the two currencies can be converted or influence each other. Many dual-currency games keep them **non-convertible** to preserve distinct loops (e.g. in Clash of Clans you generally can't turn gold into elixir directly). This segregation maintains the need to engage in both activities. On the other hand, limited conversion or trade (perhaps via a marketplace or an exchange rate) could provide flexibility but risks one currency devaluing the other. A safe approach is to **segregate primary uses** (Sporeprint only upgrades avatars, Nutrient only expands coils) while offering small crossover benefits. For instance, maybe completing others' levels yields a tiny amount of Sporeprint too, as a bonus, so pure players still trickle progress on avatar customization – but the main way to get Sporeprint remains creating good levels. Such subtle crossover can prevent players from feeling “hard-stuck” if, say, they don't enjoy level-building, without undermining the incentive to create content.
- **Parallel Progression Paths:** Balance also means ensuring that spending of each currency yields meaningful progress at similar **time scales**. If expanding one's Coil (Nutrient sink) takes vastly longer or provides a greater advantage than avatar upgrades (Sporeprint sink), the resource values will diverge. One technique is to design **complementary progression**: for example, a bigger Coil (from Nutrients) could allow hosting more or larger levels, which in turn helps you earn more Sporeprints; meanwhile avatar upgrades (from Sporeprints) might improve your crawler's abilities, helping you tackle harder levels to earn Nutrients faster. In this way, the two progression tracks feed into each other in a **positive feedback loop** – but carefully cap the extremes to avoid runaway effects. Regular analysis of resource flow is crucial: Identify if players are bottlenecked by one currency (indicating one loop is too slow) or if a resource is stockpiling without use (indicating imbalance in sinks vs. sources). As a general rule, if players find it *too hard* to generate one of the currencies, the grind becomes off-putting and they may quit; if it's *too easy*, the resource's value diminishes and the game may become trivial ². A balanced design hits the sweet spot where **earning each resource feels rewarding but not effortless**, and both creation and play activities remain worthwhile.
- **Example:** *Mario Maker 2* doesn't have a dual currency, but it offers an analogous balance in its Maker Points (earned from others liking/playing your levels) versus clearing levels (which yields completion medals). The system incentivizes both content creation and playing. In **Verdant Coil**, Sporeprint accumulation will motivate creators to design engaging levels that others want to play, while Nutrient accumulation drives players to explore and beat levels. The economy design should celebrate **both** types of participation. A possible failure mode is if a minority of players only create popular levels and hoard Sporeprints while others mostly play; to mitigate inequality, you might implement **diminishing returns or caps** on Sporeprint from any single level (ensuring a creator who pumps out many trivial levels can't infinitely farm Sporeprints). Likewise, if someone only plays but never creates, they'll have Nutrients but no Sporeprints – the game could encourage level creation by, for example, requiring at least some Sporeprint investment to unlock higher-tier avatar abilities, nudging players to try designing levels to earn those. Ultimately, maintaining a **symbiotic relationship** between the two resources keeps the community vibrant: creators need players, and players need creators, and both should advance together.

Static vs. Dynamic Reward Structures

One fundamental design decision is whether the rewards for completing levels (and for having your level completed) are **static** – fixed amounts that don’t change – or **dynamic** – varying based on conditions like difficulty, repetition, or player performance. Each approach has pros and cons:

Reward Model	Characteristics	Pros	Cons
Static Rewards	- Fixed, predictable payout for each completion or action
- Same reward regardless of level attempt count or relative difficulty	• Simple for players to understand and for designers to balance
• Clear expectations: players know exactly what a win yields	• Highly exploitable: players will optimize by grinding the easiest, fastest content repeatedly ³
• Doesn’t account for challenge or skill – completing a trivial level yields the same as a masterpiece, which can feel unfair
Dynamic Rewards	- Payout adjusts based on context, e.g. first-time completions give more, repeats give less; harder or longer levels give more; performance (speed, skill) influences reward	• Reduces farming exploits via diminishing returns (repeat plays yield lower rewards) ⁴
• Rewards skill and variety: encourages playing new or more challenging content for better payouts, keeping gameplay fresh and engaging ⁵	• More complex to design and communicate (players must understand why a reward varied)
• If tuned poorly, can feel punitive – e.g. skilled players might feel “punished” with fewer rewards when replaying for fun, or new players might be confused by fluctuating gains

In practice, most modern games use a blend: **static base rewards** with **dynamic modifiers** layered on top. For example, a level might always give 50 Nutrients on completion, but the game could grant a +100% bonus the first time you beat it, and then progressively smaller bonuses on replays (to discourage infinite farming). **First-clear rewards** or daily bonuses are common techniques: the first time (or first few times) you do something, you get full rewards, but subsequent completions yield less. This maintains an incentive to seek out new content instead of repeating a single optimal “farm” level endlessly.

Dynamic scaling can also tie to **content difficulty or player skill**. Harder levels (perhaps measured by average failure rate or a rating) might award proportionally more Sporeprint/Nutrient to compensate the effort. This gives creators an incentive to design interesting, challenging levels (not just easy farms) and players a reason to tackle tougher coils. However, automatically assessing “difficulty” can be tricky – player skill varies, and creators might try to game the system (e.g. by adding fake difficulty or tedious length to inflate rewards). A solution is to use community-driven metrics: if only skilled players beat a level and many fail, it’s likely hard and could be rewarded accordingly, whereas levels that everyone beats easily might be marked as easy and yield less. Many RPGs use an analogous system by reducing XP/loot from enemies far below your level (preventing high-level players from farming beginner zones). In a user-generated context, **dynamic difficulty adjustment** could also occur per player: if a player is repeatedly beating content

effortlessly, the game could quietly scale up the challenge or scale down rewards until they tackle content commensurate with their skill ⁶ ⁷ . This approach, as noted in *Julian Raiden's* analysis of play-to-earn games, keeps skilled players from endlessly exploiting low-risk content by **making easy tasks less lucrative over time** ⁸ ⁹ .

Verdant Coil Implementation: A balanced approach might be: each level completion gives a base Nutrient reward to the player and a base Sporeprint reward to the creator. Then implement **diminishing returns** on *repeat plays*: if you run the exact same level multiple times, rewards drop sharply after the first success. This prevents two friends from farming one trivial level back-and-forth for unlimited gain. Additionally, consider a **daily rotation or quest system** that pushes players to explore different coils (e.g. “Complete 5 new levels you’ve never beaten for bonus Nutrients”). Such systems, analogous to rotating dungeon bonuses or resource node resets in other games, ensure variety. Dynamic rewards based on coil complexity could use a formula or tier system – e.g., levels tagged “medium” difficulty by the system offer 1.5× rewards, “hard” offer 2× – but only if those tags are reliable (perhaps derived from performance data of many players). One innovative idea is **community calibration**: if a level’s clear rate is, say, under 20%, it might be labeled “Heroic” and grant an achievement or extra currency for those who beat it. This both motivates skilled players and avoids over-rewarding trivial content.

A caution: Traditional anti-farming mechanisms like **strict cooldowns or caps** can feel artificial if overused ⁹ . Players might chafe at hard limits (“You can only earn 100 Nutrients per day from playing” could feel like a forced halt). Whenever possible, **soft limits** (diminishing returns, gently scaling down outputs) are preferable to hard stops, because they **nudge** players away from repetitive grinding without completely blocking their freedom. For instance, *Guild Wars 2* famously implemented a “diminishing returns” system in certain farmed events: players noticed that doing the exact same event repeatedly started yielding less loot over time – a subtle deterrent rather than an explicit ban. Verdant Coil could use a similar philosophy: **the game should naturally encourage a healthy play pattern** (variety and challenge) via its reward curve. If a player tries to game the system, they simply find it’s not very profitable, and thus are guided back to the intended diverse experience.

Preventing Inflation, Farming, and Exploits

In any game economy, especially one with user-generated content, **uncontrolled resource inflation** and exploitative farming can quickly undermine the experience. Players are adept at finding the path of least resistance – e.g. creating a trivial level that completes itself, using alternate accounts or friends to repeatedly play each other’s levels for fast currency, or replaying an easy level hundreds of times. These behaviors can wreck balance by flooding the economy with unearned resources, trivializing progression, and demotivating honest players. To uphold long-term sustainability, designers must incorporate **“faucet and drain” balancing, anti-exploit rules, and psychological incentives** to play fair.

Faucets and Sinks: As described in economy design theory, every currency source (**faucet**) should be countered by a sink (a way to spend or remove that currency) to avoid overflow ¹⁰ . If Nutrients or Sporeprints pour in faster than they are spent, inflation occurs – players accumulate mountains of currency with nothing meaningful to buy, leading to “mudflation” (a term from MMOs for devalued currency) and boredom ¹¹ ¹² . To prevent this, calibrate the **earn rate vs. spend rate** so that players always have desirable things to spend resources on, creating a **constant demand** for the currencies. For example, if average players earn ~100 Nutrients a day, ensure that expanding or upgrading the Coil consumes a comparable amount over the same period (with top players maybe able to spend even more). If too much

currency is generated (faucet too strong), tighten it or add new sinks; if players are starved (faucet too weak), increase rewards or lower costs ¹³ ¹⁴ . The goal is a steady state where **resources flow but don't flood** – as one designer puts it, offer “just enough in-game resources to make the sinks viable and attractive, but keep them elusive enough to motivate continued play” ¹⁵ ¹⁶ .

Diminishing Returns & Scaling: We touched on dynamic rewards for repeat plays – that is a prime anti-farming measure. By scaling down rewards for highly repetitive actions, you cut off the incentive for players to self-farm trivial levels. Another approach is **server-side monitoring** and adjustment: for example, the game could track the completion frequency of each user-generated level. If one level is being completed abnormally often (e.g. one player replays their own micro-level 50 times), the system could flag it and reduce or zero out rewards from that level. This is akin to *Medium's* suggestion of scaling quest rewards inversely with completion frequency and adjusting drop rates if a particular activity is being spammed ⁵ . Some play-to-earn games even adjust **global difficulty or rewards based on economic conditions** – e.g. if too many tokens are entering the economy, the game ramps up difficulty or lowers rewards to stabilize the currency value ¹⁷ . Verdant Coil might not need such heavy-handed global scaling in a small community, but it could employ **dynamic difficulty or AI “ghost” opponents** to make content less trivial if someone is breezing through it with suspicious ease.

Exploit Prevention Rules: It's wise to establish some **hard rules** in the design to preempt obvious exploits:

- *No self-farming:* A creator should **not receive Sporeprint when they themselves (or possibly their direct IP-sharing account) complete their own level**. This encourages genuine sharing – you only profit when others play your content. (Verdant Coil likely already assumes this, since Sporeprint is earned “when other players complete your level.”)
- *Unique rewards per level/player:* Consider giving full rewards only for the **first clear** of a level per player. Replaying the same level by the same player might yield drastically reduced Nutrient (or even zero after a point). This means a creator can't have one friend beat their level 100 times to farm Sporeprints – the friend's first win gives the creator Sporeprint, but subsequent wins either give none or a token amount.
- *Quality thresholds:* To discourage “trivial levels for max gain,” the game could enforce minimal criteria for a level to be rewarded. For example, **“level must be at least 30 seconds long”** or must include a certain number of elements before it qualifies for reward payouts. (Be cautious: clever players might still make an easy 30-second level. But at least it sets a bar.) Another tactic is a **community rating system**: if players can rate levels and a level is overwhelmingly flagged as “too easy” or low-effort, the system could exclude it from reward generation or reduce its weight. This needs robust anti-abuse measures, though, to prevent mass down-voting wars.
- *Account linking and bot detection:* In online communities like Roblox, exploiters use dozens of alt accounts to farm rewards ¹⁸ . If Verdant Coil is a closed educational project, this may be less of an issue, but it's worth considering limitations such as **one account per person** policies, monitoring for multiple accounts from the same device or IP engaging in suspicious trading, and employing captcha or bot-detection for repetitive play patterns ¹⁹ ²⁰ . Modern games often have automated systems to detect when an account is behaving non-humanly (e.g. running the same route every 10 seconds for hours) and will intervene or throttle rewards in those cases.
- *Content moderation:* Because players create levels, there's the risk of **inappropriate or nonsensical content** which, aside from moral/content concerns, could be used to cheat the economy (e.g. a level that instantly completes itself to farm rewards). Having a review process or at least requiring that **creators must beat their own level before publishing** (a la *Super Mario Maker* upload requirements) helps ensure levels are legitimately completable and not pure farming scripts. It doesn't eliminate trivial “hold right to win” style levels, but it adds a bit of friction to creating junk levels.

Feedback to Players: It's important to clearly communicate these anti-exploit measures to avoid confusion. If a player's tenth run of a level gives 0 Nutrients, the UI should indicate "No reward – try other levels for more Nutrients!" rather than leaving them baffled. Transparency (to a degree) will make the systems feel **fair, not punitive**. Indeed, if dynamic adjustments are too opaque, players might feel arbitrarily punished. Strive to **frame it as encouragement**: e.g., a message after replaying a level multiple times could say "You've mastered this level! Try a new challenge for full rewards." This positive messaging aligns with the idea that the game is guiding you towards fun and growth, not simply restricting you.

Finally, always be vigilant post-launch: Watch the **data**. Players will invent new exploits; the design team must iterate with patches or live tweaks (since this is an educational project, one can treat it as a learning opportunity in game balance). Encouraging a culture of fair play and creative challenge – possibly through community events or spotlights on great levels rather than simply who farmed the most currency – can also organically steer players away from degenerate play patterns.

Engaging and Fair Resource Sinks

A healthy game economy not only needs faucets that give out resources, but also **resource sinks** – compelling ways for players to spend those resources. Sinks serve multiple purposes: they drain excess currency to prevent inflation, provide goals for players to strive toward, and let players express achievement or personalize their experience. The challenge is to design sinks that are **engaging, fair, and meaningful** – especially for top players who might accumulate large stores of currency – without feeling punitive or mandatory.

Types of Resource Sinks: In game economy design, sinks generally fall into a few categories ²¹ ²² : - *Fixed Sinks*: One-time costs that permanently unlock something. (In Verdant Coil, buying a specific avatar outfit or a coil segment expansion could be a fixed sink – you spend X Sporeprints once to get the item forever.) These are straightforward and **satisfying** because they give a clear sense of progression (e.g. "I saved up and bought the Mega Mushroom cosmetic!"). They also allow easy balancing; designers can enumerate all fixed sinks to ensure the total currency needed roughly matches what the game hands out over its lifespan ²³ . The risk is once players buy all fixed upgrades, they have **no further use** for currency. - *Repeatable Sinks*: Costs that can be paid multiple times, often for consumables or iterative upgrades. For instance, spending Nutrients to run a temporary buff, or Sporeprints to craft a new level element each time. These sinks are **linear** and limited by how much resource the player can generate (you can buy as many health potions as you can afford, for example). They ensure that even if you have excess currency, you can always convert it into some advantage or utility. In Verdant Coil, an example might be allowing players to spend Nutrients to **decorate their Coil habitat** with plants or cosmetic flourishes – each decoration costs some amount, and players can keep customizing as long as they have currency. It's not mandatory for gameplay, but it provides an endless outlet for creative expression (thus engaging top players who love to tinker). Repeatable sinks naturally scale with players' wealth ²⁴ ²⁵ . - *Exponential Sinks*: These are **scaling costs** – each new purchase or upgrade costs exponentially more than the last. This design is brilliant for endgame longevity: no matter how wealthy a player gets, there's always a higher tier they could reach by spending a lot more. Many RPGs implement this in leveling or skill upgrades (each level requires much more XP than the previous) ²⁶ . For Verdant Coil, expanding your Coil could be an exponential sink: e.g., to add the 5th room to your Coil might cost 100 Nutrients, the 6th costs 200, the 7th 400, etc. This **soaks up excess Nutrients** from the most hardcore players without affecting beginners (who are fine with just a few expansions). Exponential sinks inherently counteract inflation by matching high resource generation with high costs ²⁷ . The caveat is they can feel grindy if the increments don't provide proportional value. It's

important that each expansion or upgrade, while pricier, also **feels rewarding** (new capabilities, status, etc.), otherwise players might not bother. - **Competitive Sinks**: These are sinks where players pour resources in **competitions or leaderboards** for status, rather than direct in-game power ²⁸. For example, a leaderboard event might reward the top contributors who spent the most of a resource on some community goal. This has **no upper limit** (whales can spend endlessly), thus eliminating currency from the economy effectively ²⁹. However, it only appeals to a subset of players (often a tiny % who are extremely competitive or have surplus resources), and in a non-monetized educational game, we likely don't want to encourage an arms race of spending. Competitive sinks are more relevant in games with large communities and possibly monetization (to drive purchases), which is not our focus. So Verdant Coil might avoid this, or implement it in a gentle form like friendly competitions or puzzle contests that consume some resources to enter, but mainly for fun rather than serious loss.

For Verdant Coil, likely **fixed and repeatable sinks** for mid-level progression and **exponential sinks** for endgame will be most useful: - **Avatar customization and upgrades (Sporeprint sink)**: This should be a joyous sink, not a chore. Players spend Sporeprints to unlock new avatar cosmetics, skins, titles, or minor ability tweaks. To keep it fair, these should be mostly **cosmetic or convenience**, not outright power. Since Sporeprints reflect community appreciation of your levels, spending them on flaunting your creative prestige (cool costumes, profile badges) makes sense. Top creators thus have a way to show off without breaking the game. It's important these upgrades don't feel "mandatory" – a new player with few Sporeprints shouldn't feel at a severe disadvantage. Thus, **cosmetic sinks** are ideal: they provide long-term goals (collect all avatar pieces) and status without gating core gameplay. As GameRefinery notes, collection mechanics tap into players' satisfaction from long-term goals and "showboating to others" (displaying rare items) ³⁰ ³¹. This motivation is strong in top players; giving them a collection sink for Sporeprints keeps them engaged trying to "catch 'em all." We can design some *ultra-rare cosmetic items* priced exorbitantly in Sporeprints – a true badge of honor for veteran players who've had many people play their levels. This functions as an exponential sink in effect: only the top few will ever get enough Sporeprint for those items, but those players then have something to do with their huge reserves (preventing a scenario where they have "maxed out" and get bored). **Key is making it optional and prestigious**, not needed for progression (so it's not punitive). - **Coil expansion and level-building tools (Nutrient sink)**: Nutrient should primarily sink into expanding and enhancing one's Coil (i.e., the personal dungeon/garden that houses your created levels). For example, spending Nutrients could unlock new level slots, larger map sizes, new trap or puzzle elements to place, or decorative upgrades for your Coil environment. This serves two purposes: it's a **motivation to play others' levels** (to earn Nutrient and unlock more creative potential for your own levels) and a **direct sink** that fuels more content creation. To keep it fair, the basic ability to create fun levels should not be behind huge paywalls – early Coil expansions should be affordable (so new creators can get going quickly). But higher-tier expansions (e.g., a 10th level slot, or a special biome theme for your levels) can be quite costly in Nutrient, acting as long-term goals. This again is akin to exponential sink: "I can add one more super-level to my Coil but it costs as much as the first 5 combined." Dedicated players will save up for it, which drains their large reserves, while casual players don't *need* it if they're satisfied with a smaller coil. Also, consider **consumable sinks**: maybe Nutrient can be spent to accelerate something (though avoid the feeling of "mandatory potions"). Since we're avoiding monetization, we likely won't include "pay Nutrient to skip wait times" in a heavy-handed way, but minor conveniences could exist (e.g., spend some Nutrient to instantly "promote" your level to more players in the discovery queue, essentially advertising it; top creators might do this to attract players, spending down their Nutrient – this is just one idea that acts as a sink benefiting those who really care about exposure).

No Punitive Taxes: A common mistake in some games is to introduce sinks that feel like **upkeep or taxes**, which can sour the experience. For example, requiring players to spend resources just to maintain current status – such as durability repairs, or your Coil slowly consuming Nutrients over time – would likely feel punitive in an educational, fun-focused game. We should avoid any sink that *forces* players to continuously pay just to not lose progress. That tends to feel like a punishment and can drive players away. Instead, **every sink should feel like a positive opportunity**: you *want* to spend your currency because you get something cool or useful, not because you’ll be penalized if you don’t. For top players, resource sinks must come off as **voluntary elite challenges or vanity projects**, not chores. If a top creator amasses 1000 Sporeprints, we give them optional high-end sinks: maybe a “Master Architect” avatar costume for 1000 Sporeprints, or the ability to launch a special event level. If they choose not to spend, that’s fine – they aren’t penalized – but most will because the offerings are enticing and serve as a way to channel their success into further enjoyment.

Feedback and Progression in Sinks: It’s important that spending resources feels rewarding. When a player uses a sink, give them **clear feedback** on the benefit. For instance, if they expand their Coil, show the new space unlocked with a celebratory animation (reinforcing why it was worth 500 Nutrients). Psychologically, people are more willing to part with currency if the result is **immediately gratifying and visible**. This taps into the core drive of *ownership*: players feel invested in what they build or customize ³². A well-designed sink thus not only balances the economy but deepens a player’s attachment to the game (their upgraded avatar or elaborate Coil becomes something they’re proud of, which in turn motivates them to keep playing to utilize those upgrades).

Finally, monitor the economy: If top players still accumulate currency with nothing to spend on, be ready to introduce new sinks (perhaps new tiers of upgrades or special seasonal content). A live game (even a non-commercial one) can evolve its economy. The Lostgarden analysis emphasizes ensuring “there is a strong enough set of anchors and associated sinks to consistently pull resources up from all nodes along the chain” ³³ – meaning at every stage of progression, there should be something enticing to spend on so resources keep flowing. If Verdant Coil’s elite players ever reach a point of **overabundance with no desire to spend** (e.g., “I have all avatar items and my Coil is maxed out, now I’m just stockpiling Nutrients”), that’s a sign to introduce either *new cosmetic tiers*, *prestige levels*, or other sinks to re-engage them. For instance, maybe allow them to **prestige reset** their Coil after maxing, in exchange for a special badge, which then lets them progress again (a typical infinite progression tactic).

In summary, **meaningful sinks** maintain the value of resources and give players long-term goals. By focusing on fun, optional, and status-oriented sinks (instead of maintenance fees), we ensure that spending resources feels like a gratifying part of the game rather than a grind. This keeps both average and hardcore players invested in the loop.

Supporting Newcomers and Veterans (Progression Pacing & Difficulty)

An asynchronous, user-driven game must cater to a broad range of player skill and commitment. Beginners should be able to ramp up quickly and feel welcomed, while advanced players need ongoing challenge and satisfaction. Achieving this means implementing **scalable systems and smart matchmaking** so that each player finds a “flow” state of play – not overwhelmed, but not bored ³⁴ ³⁵.

Onboarding New Players: For novices, the early game should be **generous and tutorialized**: - Provide initial **starter rewards** – e.g. a bundle of Nutrients and Sporeprints from a simple tutorial level – so they can immediately purchase a first Coil expansion and avatar item. This gives a sense of progress within the first few minutes of play (critical for hooking interest). Many games use front-loaded rewards to ensure the first session feels rewarding; later, the pacing can normalize. Avoid any feeling of slog in the first hour. - Gradually unlock features: Perhaps new players start with the ability to play levels and earn Nutrients, but the level editor (to create their own Coil level) unlocks after a short introduction. This way they are not overwhelmed with all the tools at once. *LittleBigPlanet* and other UGC games often gate creation tools behind initial gameplay to teach the player step by step. Verdant Coil might introduce the level creation mode once a player has completed, say, three levels made by others, thereby showing them examples first. - **Difficulty gating:** New players should encounter easier coils first. This can be done via a rating system or algorithm that surfaces beginner-friendly levels to newbies. For instance, levels could have difficulty tags, or there could be an “easy mode queue” that the system directs low-skill or low-level players to. Alternatively, a skill-based matchmaking can analyze a player’s early performance (do they struggle or sail through levels?) and then recommend content accordingly. Ensuring early success (without it being a cakewalk) is important to build confidence. If a new player immediately faces brutally hard user levels, they may get frustrated and leave (the **anxiety zone** outside the flow channel ³⁶). - **Mentorship and examples:** Perhaps highlight some “dev-picked” simple levels as recommended for beginners, to set a baseline. Also, provide sample coil layouts or templates in the editor for newbies to modify, lowering the barrier to creating their first level.

Ramp-Up Mechanics: Beyond the very beginning, **progression systems** should allow beginners to catch up and feel competitive in a reasonable time. Consider **daily challenges or quests** that give extra rewards to lower-level players, effectively acting as a catch-up. For example, maybe a new player gets a series of “新人 Missions” that reward chunks of Sporeprint and Nutrient for completing basic tasks (build your first level, complete 5 levels, get your first like from another player, etc.). These one-time missions can bootstrap their progression and teach good habits. The idea is to avoid a stark gulf where veterans are so far ahead that new players feel they can never reach that status. Since this project isn’t about monetization or a zero-sum competition, we can be generous with helping newcomers reach a fun equilibrium.

Challenge and Mastery for Veterans: For experienced players, boredom is the enemy. We must provide **ongoing mastery paths** and *increased challenge*. Some strategies: - **Tiered Content or Leagues:** We could establish tiers of levels (say Tier 1, Tier 2, Tier 3) analogous to difficulty leagues. As players gain experience (or as their Coil size/Sporeprint total grows), they unlock higher tiers where levels are more complex and the stakes (rewards) are higher. This parallels how many games unlock “Nightmare” or “Heroic” modes for advanced players. In Verdant Coil, perhaps a veteran can access an “Expert Dungeon” category of user levels – these might be curated or require that the creator spent some Sporeprint to publish at that tier (ensuring only serious content appears). Advanced players will relish levels that truly test their optimized crawler builds and puzzle-solving skills. - **Advanced Tools for Creators:** On the flip side, as players become veteran level designers, we should give them more sophisticated tools or options in the level editor. For example, a newbie can make basic layouts; a veteran unlocks logic gates, secret triggers, or boss creatures for their levels. This offers a **mastery path for creation**: learning to design good levels is itself a skill progression. It keeps advanced creators engaged because they have new “toys” to master and their levels can become ever-more inventive. We must be careful: advanced tools should not strictly *better* in a way that invalidates simple levels, but rather add depth or complexity. This way, beginners aren’t punished for lack of tools, but experts have more to chew on. - **Leaderboards and Achievements:** Even without monetization, healthy competition can drive engagement. Implement leaderboards that recognize various accomplishments: e.g. “Top adventurers this week” (most Nutrients earned, or most difficult levels beaten),

“Top creators” (most Sporeprints earned from players this week, or highest-rated levels). Make sure these leaderboards reset periodically or have seasons, so new players have a chance to climb and established players have an ongoing goal to maintain status. Achievements (badges) for feats like “Complete 50 levels” or “Your level was played 100 times” give long-term objectives. The key is **recognition of mastery** – players who pour effort into the game get visible badges, titles, or other honors. According to motivational research, *progress and mastery indicators* sustain engagement by showing players how far they’ve come and what they’ve achieved ³⁷. A top player seeing a “Grandmaster Designer” title by their name (because they earned, say, 10,000 Sporeprints total) will feel a sense of pride and motivation to continue.

- **Difficulty Scaling and Flow:** We discussed dynamic difficulty in rewards; it also applies to keeping veterans challenged. The game can subtly increase challenge for players who consistently outperform content. One approach is adaptive AI in the levels (if applicable) or simply guiding them to higher-difficulty user levels. Since content is user-made, ensuring a supply of hard levels is important. The system might promote particularly challenging levels to veteran players. If for some reason the user-generated pool isn’t providing enough difficulty, the developers might occasionally create “devilishly hard official levels” as content drops for experts. These could be purely optional, bragging-rights challenges (with special cosmetic rewards for completion).

- **Continuous Skill Development:** Encourage veterans to not just grind, but refine their skills. Mastery paths could include learning to speedrun levels (maybe time-based secondary goals), optimize their Coil design to trap other players (if there’s a notion of other players trying to beat your Coil competitively), or engage in meta-strategies like designing puzzles that maximize Sporeprint yield. If the game has a social or feedback component (players can comment on levels or follow creators), veterans can also take on mentorship roles – e.g., a known good creator might help a newer one improve their level design. This fosters community and keeps experienced players invested not just in their own progress but in the game’s ecosystem.

Preserving Fun and Fairness: One risk with catering to advanced players is accidentally creating an elite clique and alienating newcomers. We must ensure that any competitive elements remain **friendly and cooperative** at heart. Since Verdant Coil’s theme is more collaborative (sharing levels) than head-to-head PvP, the tone should stay positive. Emphasize *personal bests* and *cooperative goals* over direct player vs player rankings when possible. For instance, a “community coil” event could have all players contribute levels or resources to unlock a global reward – top contributors are recognized, but everyone benefits. This drives engagement for veterans without making beginners feel worthless.

From a *flow* perspective, always aim to keep each player in that sweet spot between boredom and anxiety ³⁶. Beginners are closer to the boredom side because tasks are too easy or unclear – we fix that by giving them clear goals and easy early wins. Advanced players risk boredom because they’ve mastered the tasks – we fix that by increasing challenge and variety. If an advanced player ever feels there’s nothing left to strive for, they’ll leave; thus, the design should always have another layer (this is where live ops and new content come in, though for an educational project, “new content” might just mean the community’s output). Conversely, if a beginner feels the game is too complex or unforgiving, they’ll leave; thus, careful onboarding and a supportive community (perhaps an in-game **help system or forum** for level design tips) will help retain them.

In summary, design the progression such that **it’s a journey**: easy to start, hard to master. Use *adaptive challenges*, *tiered unlocks*, and *recognition* to make both newbies and experts feel the game is meeting them at their level. By doing so, we accommodate different skill levels and maintain long-term engagement across the player spectrum.

Psychological Motivators for Long-Term Engagement

Beyond the nuts and bolts of economy tuning, successful game progression taps into fundamental **psychological motivators**. Since this project forgoes monetization, keeping players engaged relies on intrinsic enjoyment factors: **progress feedback, variety, mastery, and collection** (as mentioned), as well as social relatedness and creativity. We will examine how Verdant Coil can leverage these to enhance fun and retention:

- **Progress Feedback:** Humans crave a sense of progress and feedback on their actions. In game design, providing clear, tangible feedback that a player is advancing is critical to satisfaction ³⁷. Verdant Coil should visually and mechanically emphasize progression. This can be done through **experience levels or ranks** (even if they're not tied to power, a player level number that increases as you play gives a constant goal), **progress bars** (show how close you are to the next Coil expansion or next avatar unlock), and **instant feedback** when goals are achieved (fireworks when you finish a level, a summary screen showing Nutrients gained, etc.). Every session should ideally end with the player feeling "I made some progress towards a goal." Even failing a level can give partial progress (e.g. small Nutrient for reaching a checkpoint, plus a tip for next time). A **quest log or achievement list** with completion percentages can spur players to keep playing to "fill the bars." Research in gamification highlights that *visible progression indicators* (like levels, points, progress bars) are key to sustained motivation because they let users know "where they stand in relation to their goals" ³⁷. In Verdant Coil, perhaps a user profile shows at a glance: Coil Level 3 → 70% to Level 4 (which motivates another run), Sporeprint Collected 300/500 towards next avatar item, etc. Feedback should also be **immediate and positive** where possible: when someone completes your level, you could get a notification: "Player X completed your level! +5 Sporeprints" – this not only rewards but also reinforces the social aspect that your creation mattered to someone.
- **Variety and Novelty:** Variety prevents boredom. Since Verdant Coil relies on user-generated levels, ideally the content stays fresh as new levels appear daily. The system can encourage variety through **rotating challenges** ("Today's featured theme: levels with lava") or simply by ensuring the *discovery feed* of levels doesn't show the same ones repeatedly. Perhaps incorporate different **biomes or level types** so players experience different aesthetics and mechanics (puzzle levels, combat levels, maze levels, etc.). If the gameplay loop is always "play a level, get resources," adding meta-layers of variety helps: e.g. daily quests inject a different goal ("find the hidden relic in any level today for a bonus"), seasonal events change up visuals or add a special temporary mechanic, etc. Variety also extends to pacing: not every session should feel identical. The **oscillation method** in level design (alternating tense and release, challenge and reward) can apply to the overall loop ³⁸. For instance, maybe after a big Coil expansion (a long-term goal), the player spends some time just enjoying creative play or easy levels before tackling the next big objective. If players feel there's always something new to see or do (new levels to try, new ways to use their resources), they are less likely to fatigue. *Unpredictability and curiosity* is one of Yu-kai Chou's core motivators ³⁹ – incorporating a bit of the unexpected (mystery rewards, surprise community events, etc.) can maintain engagement without resorting to money incentives.
- **Mastery Paths:** We covered a lot about mastery for advanced players. Here, the concept is to give players a **sense of improving skill and increasing challenge**. Psychological studies (and Self-Determination Theory) emphasize *competence* as a fundamental intrinsic motivator – people enjoy games that make them feel skilled and that they are mastering something difficult over time.

Verdant Coil can support this by **tracking personal bests** (e.g. fastest time through a level, or how many levels you beat without a failure in a row) so players can try to improve their own records. It can also provide **hard mastery goals** such as “Complete all levels in the Fire biome” or “Design a level that at least 10 people ‘like’” – achievements that require practice and creativity. When players achieve mastery milestones, celebrate it (new title, in-game certificate, etc.). Also, provide **feedback for improvement**: maybe a replay system or ghost system where you can see how a top player beat a level that you’re stuck on – so you learn techniques. That way, the path to mastery is illuminated and satisfying. Over time, a player might go from struggling with simple levels to confidently beating the hardest ones – that journey, if properly scaffolded, yields tremendous satisfaction and long-term loyalty to the game. They associate the game with *personal growth*.

- **Collection and Autonomy:** The urge to collect and complete sets is a powerful long-term hook. Verdant Coil’s dual resources naturally lend to collection: collect avatar customizations (Sporeprint sink) and collect Coil modules or decorations (Nutrient sink). By giving players many things to collect (but all optional and for fun), you engage the **completionist mindset** without forcing it. The collection should be *visible*: e.g. a wardrobe screen shows all possible avatar outfits, with silhouettes for those not yet obtained, encouraging players to “gotta get ‘em all.” This provides a steady drive that is separate from pure skill mastery – even a less skillful player can grind to collect stuff, if that’s what they enjoy. The GameRefinery report points out that collection systems add depth and tap into the psychological satisfaction of completing long-term goals ³⁰. Crucially, in our non-monetized context, collections should not confer huge gameplay advantages (to avoid pay-to-win feelings); they’re about personalization and bragging rights. That appeals to the **ownership** drive – players love having something to call their own ³². The Coil itself is a sort of collection (of levels you’ve built). Supporting players’ **autonomy and creativity** is also key: allow them to design their Coil space, customize their profile, and perhaps even create collections of favorite levels (like playlists) to share with others. When players feel the game is a canvas for their own creativity and personal expression, they are intrinsically motivated to stick around, beyond any extrinsic rewards. Verdant Coil can provide tools for that (level editor itself is a big one – it’s inherently a motivator for those who love to create, aligning with the *Empowerment of Creativity & Feedback* drive ³²).
- **Social and Community Motivators:** While not explicitly listed in the question, it’s worth noting: asynchronous games thrive on a sense of community. Encourage positive interactions – a simple **“like” or rating system** for levels gives creators validation (a powerful motivator beyond Sporeprint alone). Perhaps a commenting system or the ability to follow favorite creators and get notified of new levels. These features increase engagement by creating social bonds and friendly competition. Knowing that *real people* are playing your levels and that you’re playing theirs adds meaning to the resource exchange. It transforms progression from a grind into a shared experience (“Alice beat my level and left a note that it was really clever – that makes me want to build more!”). According to prosocial game design research, fostering **helpful, positive interactions** (like players improving each other’s experiences) can significantly increase retention and satisfaction ⁴⁰ ⁴¹. Verdant Coil’s design – rewarding you when others succeed at your challenge – is inherently prosocial (you benefit from helping entertain others). Lean into that: maybe highlight top community contributors or have a system where players can collaborate (e.g. co-design levels or run guild-like groups that share a Coil). A strong, kind community will keep players coming back for the camaraderie and creative exchange, which is a far more sustainable motivator than any virtual currency.

In sum, by aligning the game's systems with core psychological motivators – **Competence/Mastery, Autonomy/Creativity, Relatedness, and Collection/Progress** – Verdant Coil can create a deeply engaging loop. Players will stay not because they're forced by artificial gating or payments, but because they *want* to see what's next, improve their skills, show off their creativity, and be part of an ongoing, dynamic world of player-made content. Long-term engagement flows naturally when players feel a sense of **purpose and enjoyment** in the progression systems: every new level beaten or created is not just a +1 to a stat, but a meaningful step in their personal journey and the shared world.

Conclusion

Designing the progression and economy for an asynchronous, dual-resource game like Verdant Coil is a delicate balancing act. By **balancing Sporeprint and Nutrient** generation/spending, we ensure neither creators nor players are left behind – both roles feed into each other in a healthy symbiosis. Thoughtful choices between **static and dynamic rewards** help maintain challenge and prevent easy exploitation, with dynamic systems (first-clear bonuses, difficulty scaling, diminishing returns) offering a solution to repetitive farming issues if communicated fairly. Preventing economic **inflation and farming exploits** requires a combination of system design (controlled faucets and sinks, rewards scaling, unique-completion rewards) and rule enforcement (no self-farming, detecting abuse), always with an eye to keeping the game fair and fun rather than punitive.

We introduced **resource sinks** that are fun rather than frustrating – from cosmetic collections to exponential upgrades – to continually give players goals and prevent the stagnation of a capped-out economy. These sinks, especially for the most engaged players, ensure long-term purpose for accumulating resources, turning excess into brag-worthy achievements or creative expansion instead of inflation. We've also addressed **new vs. veteran player needs**: smoothing the onboarding for newcomers with quick early progress and guidance, while providing ongoing mastery challenges and recognition for experts. This layered approach maintains the all-important **flow**: newbies build confidence, experts face meaningful tests, and neither group dominates to the detriment of the other. Everyone can find appropriate content and feel a sense of growth.

Crucially, we tie it all together with **intrinsic motivators**. A progression system is only as good as the feelings it generates. By giving frequent positive feedback, a diversity of activities and goals, clear paths to mastery, and satisfying collection elements, Verdant Coil can foster that “just one more run” excitement. Academic game design literature and industry experience both affirm that when players feel **competent, autonomous, and connected**, they engage longer and more deeply than if they're just chasing extrinsic rewards. Our design prioritizes those feelings: **fairness, enjoyment, creativity, and community** over any monetization dark patterns. This makes sense for an educational project, but it's also simply good game design: a fair, player-centric economy builds trust and long-term devotion.

By studying other games and literature, we've seen examples of what works (and what fails). Verdant Coil's design takes cues from successful UGC platforms and multiplayer economies while avoiding their pitfalls: we reward creation like *Mario Maker* but pair it with a meaningful economy loop, we include multiple currencies like *Clash of Clans* but with cooperation instead of conflict, we use anti-grind measures like *Guild Wars 2*'s diminishing returns and *P2E* games' dynamic difficulty to keep things balanced ⁸ ⁹ . We implement sinks and faucets much like classic MMO economies, but tuned for a shorter play loop and non-commercial setting ¹⁰ ¹¹ .

In the end, the measure of success will be a vibrant game where players are **eager to create levels for others, excited to run each other's levels, and continually engaged in improving their avatar and Coil**. The economy should function almost invisibly to support that fun – when well-designed, players experience it not as an arbitrary system, but as a natural extension of their play. They feel rewarded for their time, challenged to improve, and valued in the community. By following the best practices outlined – balancing dual resources, using smart reward structures, preventing exploits, creating compelling sinks, easing new players in, and leveraging human motivators – Verdant Coil can achieve a progression loop that is **fair, sustainable, and deeply enjoyable** for the long term.

Sources:

- Raiden, J. (2025). *Dynamic Difficulty Scaling in P2E to Prevent Farming Exploits*. Medium. – Describes how static rewards lead to farming and outlines dynamic approaches to scale challenges/rewards based on performance and behavior ³ ⁵ . Emphasizes that traditional anti-farming methods (cooldowns, caps) feel artificial and that adjusting difficulty/rewards can naturally curb exploits ⁴ .
- Cook, D. (2021). *Value Chains: Faucet-and-Drain Economies*. Lostgarden. – Explains game economy loops with **sources (taps)** and **sinks**. Stresses balancing generation and destruction of resources to avoid overabundance or scarcity ¹⁰ ¹⁴ . Notes that exponentially increasing costs are a common solution to soak up abundant resources ²⁶ ⁴² and warns that without sufficient sinks, players lose long-term interest when currency becomes meaningless ⁴³ ¹² .
- *The Mighty Quest for Epic Loot* – **Game design example** of an asynchronous dungeon-building game with dual resources (gold for offense raids, life-force for defense trap building). Review notes that non-paying players “have to grind out in-game currency, gold and life force... by letting a castle's mines fill or raiding other castles,” reinforcing the need to engage in both roles over time ¹ . Its economy was considered balanced to not alienate free players, illustrating fair dual-resource management.
- GameRefinery (2021). *Attracting and Retaining Players with Collection Systems*. – Discusses how **collection mechanics** increase engagement by adding long-term goals and tapping into completionist motivations. Notes that collections provide extra motivation through “psychological satisfaction from progressing and completing long-term goals and, in many cases, showboating to others” ³⁰ . Also highlights that collections can introduce new sinks without breaking game balance (often cosmetic, avoiding pay-to-win) ⁴⁴ .
- Mambo Gamification Blog. *108 Gamification Elements and Mechanics*. – Emphasizes the importance of **progress and feedback** systems: “All user types require some kind of measure of progress or feedback to understand where they are in the game and what to expect.” ³⁷ This underlines using points, levels, and bars to keep players informed and motivated. Also references Yu-kai Chou's core drives, e.g. *Development & Accomplishment* (progress) and *Ownership & Possession* (collection) as fundamental motivators in games ³² , which are directly applicable to Verdant Coil's progression design.
- Roblox DevForum (2025). *Preventing Account Farming and Exploits*. – Community discussion highlighting common exploit tactics (many alt accounts farming, use of bots, etc.) and suggesting countermeasures like bot detection, captchas, and identifying multiple accounts on one device ¹⁹ ²⁰ . While about Roblox, the principles inform Verdant Coil's anti-cheat approach (limiting self-farming, monitoring unusual behavior).
- GameDeveloper.com. *Understanding the Flow Channel in Game Design*. – Based on J. Schell's and Mihaly Csikszentmihalyi's work on **Flow**, it explains balancing challenge and skill to avoid boredom

or anxiety ³⁴ . Reinforces our difficulty scaling for beginners vs. experts, ensuring the game adjusts so each player stays in the “flow channel” (neither overwhelmed nor under-stimulated) ³⁶ .

¹ The Mighty Quest for Epic Loot Review - PC | Cubed3

<https://www.cubed3.com/games/reviews/pc/the-mighty-quest-for-epic-loot>

² ¹⁰ ¹³ ¹⁴ ¹⁵ ¹⁶ ⁴² Game economy design in Free-to-Play games • Machinations.io

<https://machinations.io/articles/game-economy-design-free-to-play-games>

³ ⁴ ⁵ ⁶ ⁷ ⁸ ⁹ ¹⁷ Dynamic Difficulty Scaling in P2E to Prevent Farming Exploits | by Julianalricraiden | Mar, 2025 | Medium

<https://medium.com/@julianalricraiden/dynamic-difficulty-scaling-in-p2e-to-prevent-farming-exploits-ef790f19ef9c>

¹¹ ¹² ²¹ ²² ²³ ²⁴ ²⁵ ²⁶ ²⁷ ²⁸ ²⁹ ³³ ⁴³ Value chains – A method for creating and balancing faucet-and-drain game economies – Lostgarden

<https://lostgarden.com/2021/12/12/value-chains/>

¹⁸ ¹⁹ ²⁰ Preventing Account Farming and Exploits - Page 12 - Announcements - Developer Forum | Roblox

<https://devforum.roblox.com/t/preventing-account-farming-and-exploits/3360580?page=12>

³⁰ ³¹ ⁴⁴ Attracting and Retaining Players with Collection Systems - GameRefinery

<https://www.gamerefinery.com/attracting-and-retaining-players-with-collection-systems/>

³² ³⁷ ³⁹ 108 Gamification Elements and Mechanics

<https://mambo.io/blog/gamification-elements-and-mechanics>

³⁴ ³⁵ ³⁶ ³⁸ Understanding the Flow Channel in Game Design

<https://www.gamedeveloper.com/design/understanding-the-flow-channel-in-game-design>

⁴⁰ ⁴¹ Kind Games: Designing for Prosocial Multiplayer – Polaris Game Design Retreat

<https://polarisgamedesign.com/2022/kind-games-designing-for-prosocial-multiplayer/>